

Effect of sulfur on α -Al₂O₃-supported iron catalyst for Fischer–Tropsch synthesis

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ABSTRACT

The effects of sulfur modification (S/Fe = 0.001 ~ 0.05, molar ratio) on the α -Al₂O₃-supported iron catalysts for the Fischer–Tropsch synthesis (FTS) were investigated. It was found that the presence of sulfur promoted the reduction of FeO to metallic Fe during the H₂ reduction process, but decreased the rates of reduction, carburization and carbon deposition of the catalysts when CO is used as the reducing agent. Both the sulfide (S²⁻) and sulfate species (SO₄²⁻) are found in the S_xFe/ α -Al₂O₃ catalysts after H₂ reduction and FTS reaction. The sulfide species in the H₂-reduced S_xFe/ α -Al₂O₃ catalysts are located on the surface of metallic Fe particles. The intensity of H₂ desorption peak around 500 K is enhanced by a small amount of sulfur, while an excess of sulfur suppresses the H₂ adsorption. The dissociative adsorption of CO on H₂-reduced catalysts was also found to be inhibited by the presence of sulfur. The activity of the Fe/ α -Al₂O₃ catalyst can be decreased significantly by the addition of sulfur due to the inhibition of CO dissociation and thus reduce the formation of iron carbide phases under FTS reaction conditions. Resistance to sulfur poisoning of the Fe/ α -Al₂O₃ catalyst was found to improve with increasing the reaction temperature. The presence of sulfur also suppressed the formation of C₅₊ hydrocarbons and shifted the products to C₂–C₄ hydrocarbons. At the same time, the olefin to paraffin (O/P) ratio of the C₂–C₄ hydrocarbons decreased with increasing S/Fe molar ratio. These may have resulted from the increasing of the H/C ratio on the surface of sulfur modified catalyst under FTS conditions.

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1. Introduction

Fischer–Tropsch synthesis (FTS) is a crucial step in the production of clean fuels and valuable chemicals from syngas (a mixture of H₂ and CO). Syngas can be derived from various non-petroleum carbon resources, such as coal, natural gas, and biomass [1,2]. Compared with other metal catalysts, iron-based catalysts produce less methane and more olefins when the reaction is performed at temperatures higher than 573 K. Additionally, iron-based catalysts have a higher water–gas shift (WGS) activity and a higher resistance to the impurities present in syngas than the other metal catalysts, and are suitable for the CO-rich syngas derived from coal or biomass [2,3].

It is well known that the poisoning of iron-based catalysts by sulfur compounds is a critical problem in FTS to hydrocarbons from coal-derived syngas [4]. Substantial works have been carried out in the past to establish the effects of sulfur on iron-based catalysts [4–22]. However, there are also considerable disagreements and discrepancies among the studies regarding the role of sulfur in the FTS process. In many cases, sulfur will lead to a decline in activity [9,11,12,19] and olefin selectivity of the catalysts [15,20] accompanied by increasing of the selectivity of CH₄ and gaseous C₂–C₄ hydrocarbons [12,15]. On the contrary, some researchers reported that low concentrations of sulfur have positive effects on the activity [13,15,16,18] and olefin selectivity [16] of iron-based FT catalysts. It was also reported that sulfur could improve the heavier hydrocarbon selectivity [21] and reduce CH₄ production in the FTS reaction [16,22]. The effect of sulfur in the FTS process have been reported by Madon et al. [4] and McCue et al. [17] in two excellent reviews. They emphasized that the interactions of sulfur with FTS catalysts appeared to be quite complicated and that both poisoning and promoting effects had been observed. The controversies

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may have resulted from the differences in catalyst composition, the FTS reaction conditions as well as quantity and nature of the sulfur species in the catalysts. In addition to these factors, many studies on the interaction of sulfur with iron catalysts have focused on the exposure of a catalyst to a gaseous sulfur source. This would produce concentration gradients of sulfur along the length of the catalyst bed and pose severe problems in the interpretation of the effect of sulfur. At the same time, interpretation of the effect of sulfur in a catalyst can be clouded by other modifiers in the catalysts such as alkali metals (Na [13,16,23], K [5,7,15,19]). In spite of the recognized importance of the sulfur, the roles of sulfur in FTS reaction is still not well understood and remains to be explored. For this reason, a detailed study of the effect of sulfur on the iron catalysts without any other modifiers was carried out. Particular attention has been paid to surface compositions and nature of the sulfur species in the freshly prepared, reduced and used catalysts as well as the effect of sulfur on the textural, reduction and carburization properties of the catalysts. Sulfur-modified iron catalysts were prepared via impregnation of the support with a solution of iron nitrate and ammonium sulfate. Such a method allows us to get a better control over the quantity of sulfur introduced as well as a more uniform distribution of sulfur species in the catalyst bed. α - Al_2O_3 was chosen because it is an excellent support for iron catalyst in the selective production of lower olefins from syngas and allows a better interaction between the sulfur and the iron phase [16,24,25]. Catalyst with S/Fe molar ratios varying from 0.001 to 0.05 were prepared and were characterized by transmission electron microscopy (TEM), (*in situ*) X-ray diffraction (XRD), X-ray photoelectron spectroscopy (XPS), and temperature-programmed reduction/desorption (TPR/TPD), whereas the FTS reaction is conducted in a fixed-bed reactor at industrially relevant operating conditions. It is expected that the relevant studies can provide a better understanding of the nature of the interactions between sulfur and iron, and will be helpful for predicting lifetime as well as product distribution of the catalyst.

2. Experimental

2.1. Preparation of sulfur-modified Fe/ α - Al_2O_3 catalysts

The sulfur-modified Fe/ α - Al_2O_3 catalysts were prepared via the incipient wetness impregnation method. 5.0 g of α - Al_2O_3 (surface area: 8 m²/g, pore volume: 0.24 mL/g, Alfa Aesar) was impregnated with an aqueous solution (1.5 mL) of $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (3.5 g, Sinopharm Chemical Reagent, $\geq 98.5\%$) and $(\text{NH}_4)_2\text{SO}_4$ (0.0016–0.0817 g, Sinopharm Chemical Reagent, $\geq 99.0\%$). The α - Al_2O_3 support was treated in vacuum at 383 K for 12 h before use. The samples were then dried at 383 K for 12 h. The solid materials were finally heated under an air flow of 30 mL/min to 673 K (heating rate = 10 K/min) and calcined at 673 K for 2 h to obtain the oxidized catalyst precursors denoted as $\text{S}_x\text{Fe}/\alpha\text{-Al}_2\text{O}_3$, in which x represents S/Fe molar ratio. The Fe loading in all catalysts was 10 wt%. The α - Al_2O_3 support and Fe/ α - Al_2O_3 catalyst were tested by a Bruker Tiger S8 X-ray fluorescence (XRF) spectrometer. The results showed that content of sodium and sulfur in the samples was below the instrument detection limit.

2.2. Catalyst characterizations

The microstructure of Fe/ α - Al_2O_3 catalysts was examined *ex-situ* by a FEI Tecnai F30 transmission electron microscopy (TEM) operated at an accelerating voltage of 300 kV. The sample for TEM measurement was prepared by depositing a drop of Fe/ α - Al_2O_3 /ethanol suspension on a carbon-film-coated copper grid.

X-ray powder diffraction (XRD) analysis was carried out by a Panalytical X'pert PRO diffractometer scanning 2θ from 20 to 80°. Cu-K α radiation ($\lambda = 1.5406 \text{ \AA}$) obtained at 40 kV and 30 mA was used as the X-ray source. The *in situ* XRD experiment was performed on the same instrument with an Anton Paar XRD-900 reaction chamber. In the *in situ* XRD experiment, diffraction pattern of the catalyst was collected with 2θ ranging from 20 to 80°. The scanning rate was 3°/min. In each *in situ* XRD experiment, the sample was heated in flowing H_2 (99.999%, 120 mL/min) up to 873 K at a rate of 10 K/min and the XRD patterns were recorded at the prespecified temperatures.

H_2 temperature-programmed reduction (H_2 -TPR) experiment was performed on a GC-TPR apparatus. Prior to the TPR test, the fresh catalyst (0.10 g) was pretreated with a flow of $\text{O}_2/\text{Ar} = 5/95$ mixture (v/v, 30 mL/min) at 573 K for 30 min. After cooled down to 323 K, the sample was purged with a flow of $\text{H}_2/\text{Ar} = 5/95$ mixture (v/v, 50 mL/min) until the baseline was flat. The TPR profile of the catalyst was obtained by heating the sample from 323 to 1173 K at a rate of 10 K/min. The hydrogen consumption was monitored by a TCD.

CO temperature-programmed reduction (CO-TPR) experiment was performed on a TPR apparatus equipped with a Hiden Qic-20 mass spectrometer (MS). Typically, the fresh catalyst (0.10 g) was pretreated with a flow of He (99.999 %, 30 mL/min) at 673 K for 15 min and cooled down to 373 K. The sample was then purged with a CO/He = 5/95 mixture (v/v, 50 mL/min) at 373 K until the baseline was flat. CO-TPR was performed in the CO/He flow (50 mL/min) by raising the temperature to 1073 K at a rate of 10 K/min. Ions with m/z values at 28 (CO) and 44 (CO_2) were detected during on-line measurements.

H_2 temperature-programmed desorption (H_2 -TPD) experiment was performed on a TPD apparatus equipped with a Hiden Qic-20 mass spectrometer. The catalyst (0.10 g) was first reduced with a flow of pure H_2 (30 mL/min) at 673 K for 120 min. After cooled to 303 K under H_2 , the sample was purged with Ar (99.999%, 50 mL/min) at 303 K for about 30 min until the baseline leveled off. The H_2 -TPD profile of the catalyst was obtained by heating the sample from 323 to 1073 K at a rate of 10 K/min in a flow of Ar (50 mL/min). Ions with m/z values at 2 (H_2) and 18 (H_2O) were detected during the on-line measurements.

The CO-TPD experiment was performed on the same apparatus as used in the H_2 -TPD experiment. The fresh catalyst (0.10 g) was first reduced with pure H_2 (99.999%, 30 mL/min) at 673 K for 2 h. The sample was then flushed with He flow (99.999%, 50 mL/min) at 673 K for 10 min and cooled down under He to 303 K. CO adsorption was performed by switching the He flow to a CO flow (99.99 %, 30 mL/min) at 303 K and treating for 10 min. The sample was then purged with He flow for about 30 min until the baseline was flat. The CO-TPD profile of the catalyst was obtained by heating the sample from 303 to 1073 K at a rate of 10 K/min in a flow of He (50 mL/min). Ions with m/z values at 28 (CO) and 44 (CO_2) were detected during the on-line measurement.

X-ray photoelectron spectroscopy (XPS) analysis was performed on an Omicron Sphera II hemispherical electron energy analyzer with an *in situ* reaction cell attached to the instrument. Monochromatic Al K α X-ray source (1486.6 eV, anode operating at 15 kV and 300 W) was used as incident radiation. The binding energy (BE) of the element was reference to the Al 2p $_{1/2}$ photoelectron peak at 74.3 eV. This reference gave the BE values with an accuracy of ± 0.1 eV. The surface atomic ratio was estimated from the peak areas of the elements corrected by the relative sensitivity factors of the corresponding elements [26]. In each experiment, the catalyst sample in the *in situ* reaction cell was first heated under a flow of $\text{O}_2/\text{N}_2 = 21/79$ mixture (v/v, 50 mL/min) from room temperature to 673 K at a heating rate of 10 K/min and treated at 673 K for 20 min. The treated sample was then transferred under vacuum to the ana-

lytical chamber to record the XPS spectra of the catalyst in the oxidation state. After that, the catalyst was transferred again to the reaction cell, and heated at a rate of 10 K/min from room temperature to 703 K under a flow of H_2 (50 mL/min) and reduced at 703 K for 120 min. After cooling to 423 K under H_2 flow, the reduced sample was evacuated and transferred under vacuum to the analytical chamber to record the XPS spectra of the catalyst in the reduction state. Finally, the reduced catalyst was transferred to the reaction cell and switched to a flow of $\text{H}_2/\text{CO} = 1$ mixture (v/v, 10 mL/min) at 573 K to conduct the CO hydrogenation reaction. After treating with H_2/CO mixture at 573 K for 60 min, the catalyst sample was evacuated and transferred under vacuum to the analytical chamber to record the XPS spectra of the reacted catalyst.

2.3. Catalytic reaction

The Fischer–Tropsch synthesis (FTS) reaction was carried out at 2.0 MPa, 573 K and a H_2/CO ratio of 1 (v/v) in a stainless steel high-pressure fix-bed flow reactor (7.0 mm inner diameter). H_2/CO ratio of 1 is close to that for syngas derived from coal or biomass. Prior to the reaction, the catalyst was reduced by H_2 (0.10 MPa) at 673 K for 2 h (heating rate = 10 K/min). After the reduced catalyst was cooled to 473 K, a syngas flow of $\text{H}_2/\text{CO}/\text{Ar} = 48/48/4$ (v/v/v) was introduced into the reactor. The pressure in the reactor was then increased gradually from 0.10 to 2.0 MPa and the temperature was raised to the desired temperature. The products were analyzed by two on-line gas chromatographs. Ar, CO, CO_2 , CH_4 , C_2H_4 , C_2H_6 , C_3H_6 and C_3H_8 were analyzed by a SICT (Shanghai Institute of Computing Technology) GC-2000 III gas chromatograph equipped with a TCD, and a TDX-02 and a Porapak Q columns using H_2 as a carrier gas. The hydrocarbons were analyzed by a SICT GC-2000 III equipped with a FID and a PONA capillary column using He as a carrier gas. 4% Ar was used as an internal standard for calculation of CO conversion. The product selectivity was determined by a methane concentration correlation method and was calculated on a carbon basis.

3. Results and discussion

The XRD patterns of the fresh $\text{S}_x\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ catalysts with S/Fe atomic ratios between 0 and 0.05 are shown in Fig. 1. Besides diffraction peaks of $\alpha\text{-Al}_2\text{O}_3$, the diffraction peaks originating from iron oxide with $2\theta = 24.2, 33.2, 35.6, 40.9, 49.5, 54.1, 62.5, 64.0^\circ$ are also detected. The latter are similar to that of the hematite ($\alpha\text{-Fe}_2\text{O}_3$) species (JCPDS No. 01-084-0306). However, no diffraction peak belonging to a sulfur compound is seen. This may have

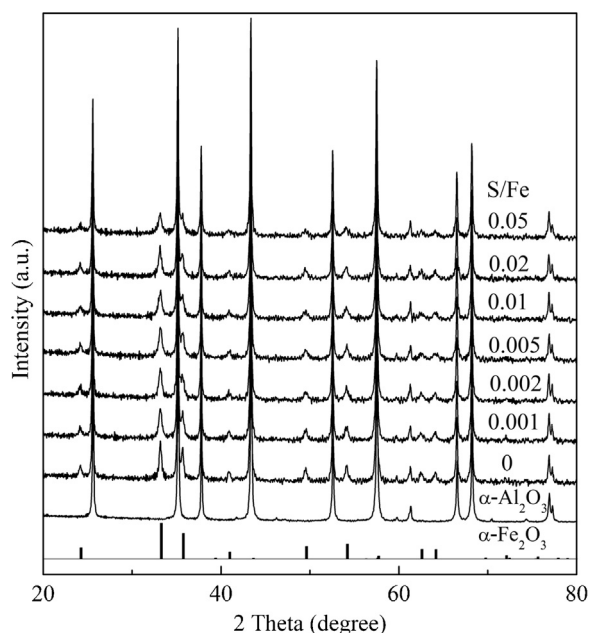


Fig. 1. XRD patterns of the as prepared $\text{S}_x\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ catalysts with different S/Fe molar ratios.

resulted from that the sulfur content in the catalysts is below the detection limit of XRD. With the increasing of sulfur content in the catalysts, the intensity of the characteristic peaks of $\alpha\text{-Fe}_2\text{O}_3$ in the catalysts has been gradually weakened. This effect may be attributed to less of crystallinity as a result of the interaction between Fe_2O_3 and sulfur.

The morphology and size of the iron species in the $\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ and $\text{S}_{0.01}\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ catalysts after H_2 reduction at 673 K was studied by TEM. As shown in Fig. 2, the iron-containing nanoparticles located at the exterior surface of $\alpha\text{-Al}_2\text{O}_3$ can be easily distinguished from the support material. Average iron (oxide) particle sizes of the $\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ and $\text{S}_{0.01}\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ catalysts are 15 ± 5 nm and 16 ± 6 nm, respectively. Both catalysts showed similar morphology.

H_2 -TPR and *in situ* XRD experiments were carried out to study the effect of sulfur on the reducibility of the $\text{S}_x\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ catalysts. As shown in Fig. 3, the H_2 -TPR profile of the sulfur-free $\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ catalyst showed three distinct reduction stages in the temperature range of 500–1100 K with temperature at peak maximum (T_m) at

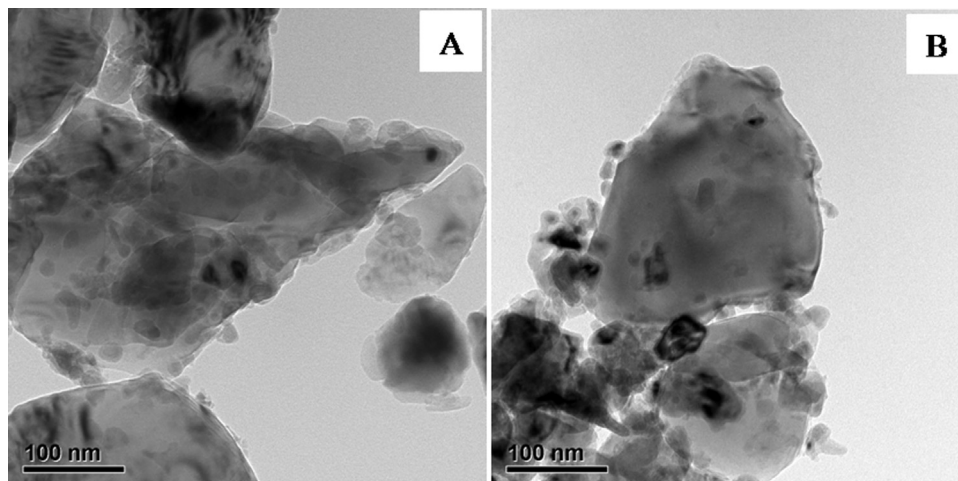


Fig. 2. TEM images of the (A) $\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ and (B) $\text{S}_{0.01}\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ catalysts after H_2 reduction at 673 K for 2 h.

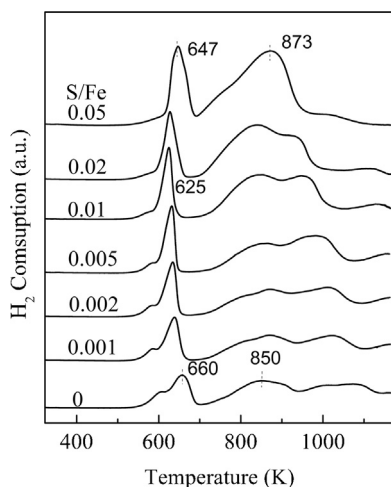


Fig. 3. H_2 -TPR profiles of the $S_x\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ catalysts with different S/Fe molar ratios.

660 K (peak 1), ~ 850 K (peak 2) and above 940 K (peak 3). From the deconvolution results of the TPR profile of the $\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ catalyst (Table S1, Supporting information), it can be seen that the ratio of the peak areas for peak 1 and peak 2 is close to the theoretical ratio of the H_2 consumption corresponding to the reduction of Fe_2O_3 to Fe_3O_4 and Fe_3O_4 to FeO , respectively. The peak with T_m above 940 K (peak 3) can be assigned to the reduction of FeO to metallic Fe. Compared with the areas of peak 1 and peak 2, the area of peak 3 (at temperatures below 1073 K) is lower than that expected for the complete reduction of FeO to metallic Fe. This may have resulted from the formation of unreducible iron aluminate as a result of strong interaction between FeO and the alumina support. When sulfur was added, the T_m for the reduction of Fe_2O_3 to Fe_3O_4 decreased from 660 to 640 K, and a further shift of T_m to lower temperature was observed as S/Fe molar ratio increased from 0.001 to 0.01. Similar phenomenon was observed in the H_2 -TPR profiles of the unsulfided and sulfide iron catalysts [13]. As shown in Fig. 3, with the increasing of S/Fe molar ratio in the catalysts from 0 to 0.05, the T_m of peak 3 also shifted towards lower temperature. This result suggests that the presence of sulfur also promoted

the reduction of FeO to metallic Fe in H_2 . A similar result was reported when sulfur modified Fe-based catalysts were reduced in H_2 [13,15]. According to the study of Li et al. [15], the higher reducibility of sulfur-modified iron-based catalysts was attributed to the decreasing of catalyst surface basicity in the presence of sulfur. Sulfur was also found to assist with oxygen removal from the iron-oxide lattice [13], and may increase the catalyst reducibility.

The XRD patterns of $\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ catalyst recorded in a flow of H_2 (120 mL/min) during stepwise heating of the sample from 298 to 873 K are shown in Fig. 4A. When the temperature of the sample was raised to 593 K, diffraction peaks of Fe_2O_3 ($2\theta = 24.2, 33.2, 35.6, 40.9, 49.5, 54.1, 62.5, 64.0^\circ$) decreased significantly in intensity, and the characteristic peaks of Fe_3O_4 phase ($2\theta = 30.1, 35.4, 43.1, 56.9, 62.5^\circ$) were observed, indicating the reduction of Fe_2O_3 to Fe_3O_4 . The XRD peaks of Fe_2O_3 were found to vanish when temperature was raised to 613 K. The diffraction peaks attributed to FeO ($2\theta = 36.1, 41.9^\circ$) and metallic Fe ($2\theta = 44.7, 65.0^\circ$) started to emerge at 653 K, revealing the transformation of Fe_3O_4 to FeO and metallic Fe. When temperature was increased to 713 K, the diffraction peaks of Fe_3O_4 disappeared and the intensity of the FeO peak went to the maximum. With a further rise in the temperature, the peak intensity of metallic Fe increased gradually. The reduction behavior of iron-based catalysts has also been investigated extensively by the Mössbauer spectroscopy [27–31]. Based on the results of *in situ* Mössbauer spectroscopy characterization on the phase transformation of a precipitated iron-based catalyst during isothermal reduction, Wang et al. found that the reduction of paramagnetic (PM) $\alpha\text{-Fe}_2\text{O}_3$ (70%) and superparamagnetic (spm) Fe^{3+} (30%) in the fresh catalyst proceed via different steps. PM $\alpha\text{-Fe}_2\text{O}_3$ is firstly reduced to magnetite and then to metallic iron, while the reduction of spm Fe^{3+} proceeds in three consecutive steps: it is first reduced to magnetite with a significantly rapid rate, then to non-stoichiometric wüstite, and finally to metallic iron [31]. In general, these results are consistent with the results of *in situ* XRD characterization on the $\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ in the present study which indicate that reduction of the Fe_2O_3 species on $\alpha\text{-Al}_2\text{O}_3$ in H_2 to metallic Fe is a three-step process (i.e. $\text{Fe}_2\text{O}_3 \rightarrow \text{Fe}_3\text{O}_4 \rightarrow \text{FeO} \rightarrow \text{metallic Fe}$).

Similar XRD patterns were observed when a $\text{S}_{0.02}\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ catalyst was heated in a H_2 flow from 473 to 873 K (Fig. 4B). However, compared to the reduction of $\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ catalyst, the

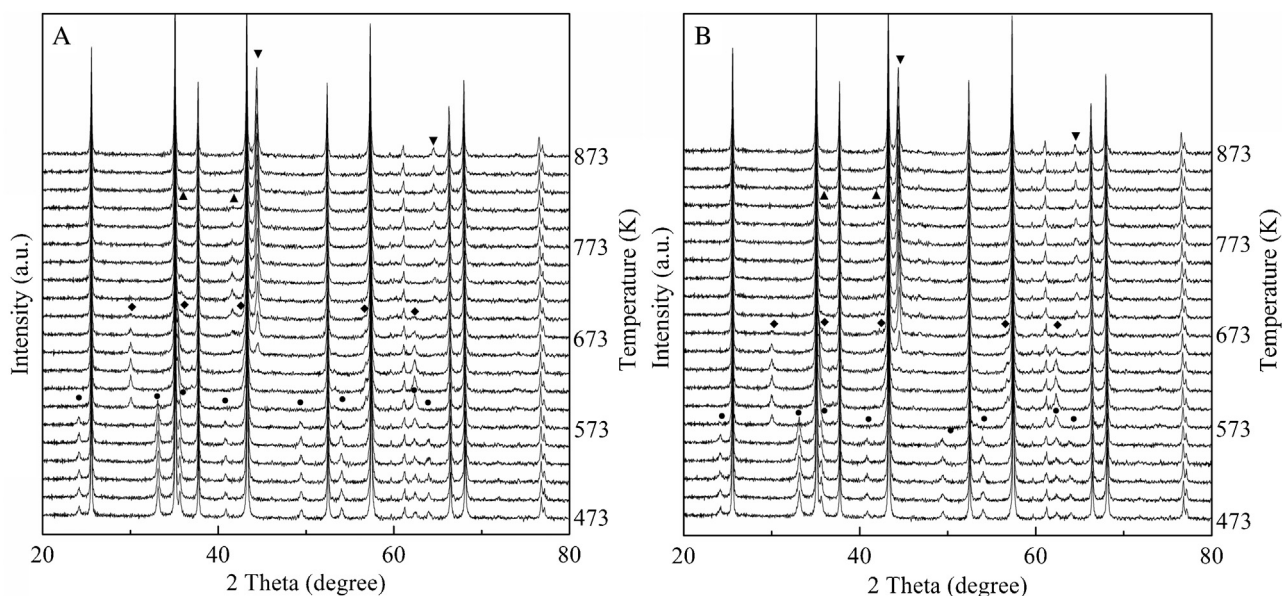


Fig. 4. *In situ* XRD patterns of (A) $\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ and (B) $\text{S}_{0.02}\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ catalyst recorded at the indicated temperature during H_2 reduction process. (● $\alpha\text{-Fe}_2\text{O}_3$ ◆ Fe_3O_4 ▲ FeO ▼ Fe).

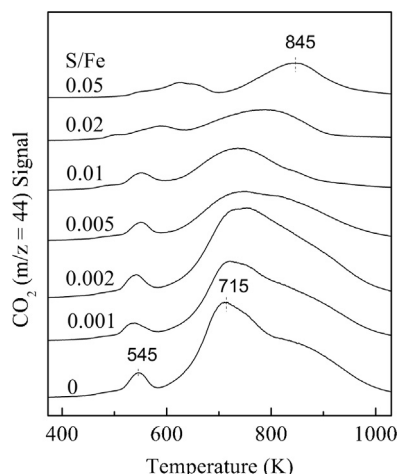


Fig. 5. CO-TPR profiles for the $S_x\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ catalysts with different S/Fe molar ratios.

reduction of Fe_2O_3 to Fe_3O_4 in $S_{0.02}\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ occurred at a lower temperature (573 K), and more metallic Fe species were formed in the $S_{0.02}\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ catalyst when reduction temperature was raised to 653 K. Another noticeable difference between the XRD pattern of $\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ and $S_{0.02}\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ catalysts is that no diffraction peak corresponding to FeO was detected in the latter during the reduction process. These results are in agreement with the results of H_2 -TPR characterizations shown in Fig. 3, and indicate that the presence of sulfur promotes the reduction of Fe_2O_3 to Fe_3O_4 as well as the reduction of FeO to metallic Fe.

CO-TPR experiments were carried out to investigate the effect of sulfur on the reduction and carburization of the Fe species in the $S_x\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ catalysts under CO atmosphere and the results are shown in Fig. 5. Two groups of CO_2 peaks are seen in the CO-TPR profile of sulfur-free $\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ catalyst. The first CO_2 formation peak observed in the temperature range of 450–570 K (peak 1) can be assigned to the reduction of Fe_2O_3 to Fe_3O_4 . Fe_3O_4 is the only Fe containing species detected by XRD in the catalyst sample after CO reduction at 573 K (Fig. S1). The CO_2 formation peak above 570 K (peak 2) can be related to the conversion of CO via the Boudouard reaction ($2\text{CO} \rightarrow \text{C} + \text{CO}_2$) as well as the reduction of Fe_3O_4 to metallic Fe and carburization of Fe_3O_4 to the mixtures of iron carbides [32]. The carburization of Fe_3O_4 to iron carbides is evidenced by the observation of a mixture of iron carbides in the XRD pattern of the sulfur-free $\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ catalyst after the CO reduction at 1073 K (Fig. S1). The assignment of peak 2 is also supported by the peak deconvolution results of the CO-TPR signals of the sulfur-free $\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ catalyst (Table S2). It can be seen that the intensity ratio (1/12) of peak 1 and peak 2 in Fig. 5 is smaller than the stoichiometry (1/8) of CO_2 production for the reduction of Fe_2O_3 to Fe_3O_4 and Fe_3O_4 to Fe by CO, suggesting that other reactions such as carburization of the iron species (Fe_3O_4 or Fe) to iron carbides and the Boudouard reaction also took place during the CO-TPR process. According to the results of Mössbauer analysis [33], a Mo-promoted Fe catalysts (100Fe5Mo) pretreated with pure CO at 280 °C were reduced to Fe_3O_4 , $\chi\text{-Fe}_5\text{C}_2$ and superparamagnetic species (Fe^{3+} and/or Fe^{2+}). The results of *in situ* TPR-EXAFS/XANES study also confirmed that the Fe/Si catalysts (100Fe:4.6Si) promoted with group I alkali metals were firstly reduced to Fe_3O_4 and then carburized to iron carbides under a CO/He flow [34]. In summary, The Fe_2O_3 species in the $\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ catalyst undergoes a consecutive stepwise reduction to Fe_3O_4 and iron carbides under CO atmosphere.

With the increasing of S/Fe molar ratio in the $S_x\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ catalysts from 0.002 to 0.05, the intensity of the second CO_2 peak in Fig. 5

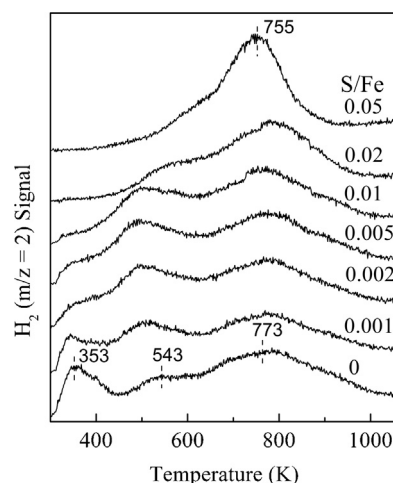


Fig. 6. H_2 -TPD profiles of the H_2 -reduced $S_x\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ catalysts with different S/Fe molar ratios.

decreases gradually. It was also found that the temperature of the two CO_2 formation peaks increases significantly with the increasing of S/Fe molar ratio from 0.01 to 0.05. Moreover, the result of XRD measurements (Fig. S1) indicated that only a metallic Fe phase was detected in the $S_{0.05}\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ sample reduced with CO at 1073 K. These results suggest that the addition of sulfur decreases the rates of reduction and carburization of the iron species in $S_x\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ catalysts by CO. Similar results were reported by Li et al. [15] in the study of sulfate promoted iron-manganese catalysts. Obviously, the reduction behavior of the $S_x\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ catalysts by CO (Fig. 5) is different from that by H_2 (Figs. 3 and 4). This may have resulted from that CO dissociation on the metallic iron surface is retarded by the presence of sulfur (as observed in CO-TPD experiment shown below) [35–37]. It was reported that the presence of sulfur on the iron surface results in the depletion of electrons from the iron [35] and/or blocking of the available CO dissociation sites and decreasing the CO dissociative adsorption [36,37]. As a result of these, the reduction, carburization and carbon deposition of the iron species in the catalysts under CO atmosphere are inhibited.

The H_2 -TPD was used to investigate the effect of sulfur on the H_2 adsorption behavior of the catalysts. As can be seen from Fig. 6, three peaks with temperature at peak maxima (T_m) at 353, 543 and 773 K were seen in the H_2 -TPD profile of $\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ catalyst. As reported in literature, hydrogen desorption usually occurs below 600 K on single-crystal iron planes [38,39] or polycrystalline iron surfaces [40–43], which is similar to the low temperature desorption peaks observed on the $\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ catalyst in the present study. The H_2 desorption peak at temperatures above 600 K can be ascribed to a strong chemisorption state of hydrogen on the catalyst [44]. It was also reported that water held by the Al_2O_3 support could be driven off at temperatures above 713 K and reacted with the metallic Fe to form a H_2 peak at above 700 K [40]. Nevertheless, it was found that the hydrogen species in a strong chemisorbed state could not react with N_2 to form ammonia at temperature below 773 K [44]. Considering the CO hydrogenation temperature in this study is below 623 K, it is rational to suggest that the hydrogen species desorbed from the $\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ catalyst at temperature below 623 K (Fig. 6) should be responsible for the CO hydrogenation reaction over the $\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ catalyst.

As illustrated in Fig. 6, the H_2 desorption peak at 353 K decreased gradually in intensity with the increasing of S/Fe molar ratio in the catalysts from 0 to 0.01, and eventually disappeared at S/Fe = 0.02. Whereas the intensity of H_2 desorption peaks at 543 K increased with the increasing of S/Fe molar ratio from 0.001 to 0.002 and remained unchanged with a further increase in the S/Fe molar ratio

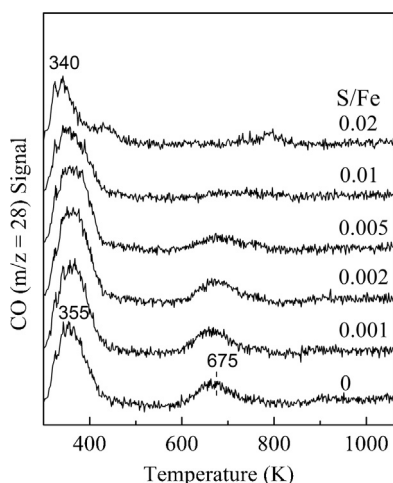


Fig. 7. CO-TPD profiles of the $S_x\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ catalysts with different S/Fe molar ratios.

from 0.005 to 0.01. Similar phenomena were observed on the industrial iron catalysts poisoned with different amounts of sulfur [45]. These results suggest that a small amount of sulfur can enhance the amount of H_2 desorption from iron catalyst surface at about 500 K. The underlying reason for the observation may be that sulfur is conducive to the reduction of Fe_2O_3 to Fe^0 in H_2 , and formation of metallic Fe species is in favor of the activation and adsorption of hydrogen on the catalyst. On the $S_{0.05}\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ catalyst, the H_2 desorption peaks at temperature below 600 K almost completely disappeared. This phenomenon can be understood in terms of blockage of the site for H_2 adsorption on the catalyst surface by sulfur at relatively high sulfur coverage. It was found that sulfur adsorption into $c(2 \times 2)$ structures on $\text{Fe}(100)$ surface completely blocked the site for H_2 adsorption at 200 K [35].

Several studies have been carried out to study the effect of sulfur on the adsorption behavior of CO over low index planes of iron metal surface [35,36,46–48], whereas the effect of sulfur on CO adsorption over supported iron catalysts is rarely studied. To shed more light on the effect of sulfur on CO adsorption and dissociation over $\alpha\text{-Al}_2\text{O}_3$ supported iron catalysts, a CO-TPD study of the H_2 -reduced $S_x\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ catalysts was conducted. The results are shown in Fig. 7. The adsorption of CO on H_2 -reduced $\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ catalyst resulted in two major CO desorption peaks in the temperature range of 300 and 1050 K. The peak at lower temperature (355 K) corresponds to the desorption of molecularly adsorbed CO, while the one at higher temperature (675 K) is ascribed to recombination of C and O of the dissociated CO [49,50]. These results are in agreement with the CO desorption profiles on Fe/SiO_2 [49] and Fe/HOPG [50] catalysts. As reported in the literature [35,51], the dissociated carbon and oxygen may have resulted from further dissociation of molecularly adsorbed CO at 303 K and higher temperature. As the S/Fe molar ratio increased (especially on the sample with S/Fe molar ratio higher than 0.005), the desorption peak of molecularly adsorbed CO shifted to lower temperature, and the desorption peak of dissociated CO decreased in intensity. On the sample with S/Fe molar ratio higher than 0.01, the desorption peak of dissociated CO disappeared completely. These results clearly showed that the addition of sulfur inhibited the dissociative adsorption of CO on H_2 -reduced $\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ catalysts, which is consistent with the trend observed on the low index planes of iron metal [35,36]. These results can be understood in terms of the bonding interaction between the Fe surface and sulfur atoms which deplete high-lying occupied Fe D-states near E_F and reduce the metal's ability to back-donate electrons into the 2π acceptor orbital of CO [35,52]. In addition, sulfur can also reduce the CO dissociation by blocking the available sites for dissociated carbon and

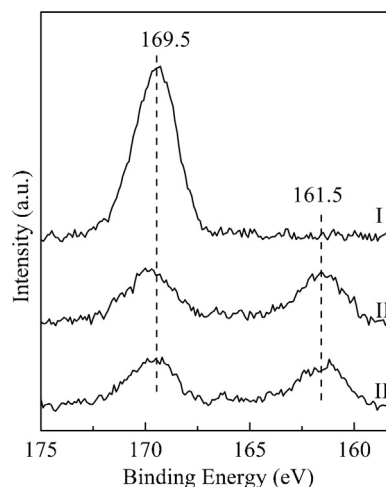


Fig. 8. S 2p XPS spectra of the $S_{0.05}\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ catalyst after (I) treating under a flow of $\text{O}_2/\text{N}_2 = 21/79$ mixture (v/v, 50 mL/min) at 673 K for 20 min, (II) reducing under a flow of H_2 (50 mL/min) at 703 K for 120 min, (III) exposure to a flow of $\text{H}_2/\text{CO} = 1$ mixture (v/v, 10 mL/min) at 573 K for 60 min.

oxygen [36]. It has been found that the dissociated carbon and oxygen atoms occupy randomly the four-fold hollow sites on $\text{Fe}(100)$ [53] while sulfur atoms occupy the same sites on $\text{Fe}(100)$ [54].

To gain further insight into the nature of the sulfur species in the $S_x\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ catalysts before and after the FTS reaction, a detail XPS analysis on the catalyst was performed. Figs. 8 and 9 show the S 2p and Fe 2p XPS spectra of a freshly prepared $S_{0.05}\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ catalyst and the samples after H_2 reduction at 703 K as well as after the FTS reaction at 573 K. The corresponding surface atomic ratios of the elements over the samples calculated based on the XPS results are given in Table 1. The single peak with B.E. at 169.5 eV observed in S 2p spectrum I indicates that the sulfur is in the form of sulfate (SO_4^{2-}) on the surface of a freshly prepared $S_{0.05}\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ catalyst [23]. It is also found that the $S_{\text{total}}/\text{Fe}$ ratio on the fresh $S_{0.05}\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ catalyst is much higher than the stoichiometry of S and Fe elements in the catalyst, indicating that the sulfate is mainly located on the catalyst surface. After reducing with H_2 at 703 K for 2 h, XPS peak intensity of sulfate decreased significantly, and a new peak with B.E. at 161.5 eV corresponding to sulfide species (S^{2-}) was observed [23,55], indicating the partial transformation of sulfate species into sulfides after H_2 reduction at 703 K. The calculated surface $\text{S}^{2-}/S_{\text{total}}$ atomic ratio reveals 45% of S^{2-} in total sulfur, indicating that the sulfide (S^{2-}) and sulfate species (SO_4^{2-}) are the main sulfur components in the H_2 -reduced $S_x\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ catalysts. Bromfield et al. studied the effect of reduction temperature on surface sulfur species over the sulfided precipitated-iron catalysts using XPS [23]. It was found that besides assisting in the conversion of sulfate to sulfide, H_2 reduction at high temperature (673 K) also caused segregation of sulfur species on the iron surface, and the diffusion of sulfur species to the surface is favored by higher reduction temperatures. From the results shown in Table 1, how-

Table 1

Surface composition of the $S_{0.05}\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ catalyst after the O_2/N_2 calcination at 673 K, H_2 reduction at 703 K and FTS reaction at 573 K.

Pretreatments	Atomic ratios calculated from XPS				
	Fe/Al	$S_{\text{total}}/\text{Al}^a$	$S_{\text{total}}/\text{Fe}$	$\text{S}^{2-}/\text{Fe}^b$	$\text{S}^{2-}/S_{\text{total}}$
O_2/N_2 , 673 K	0.26	0.052	0.20	0	0
H_2 , 703 K	0.19	0.035	0.18	0.083	0.45
H_2/CO , 573 K	0.18	0.034	0.18	0.086	0.47

^a S_{total} : total amount of sulfur species.

^b S^{2-} : the amount of sulfide species.

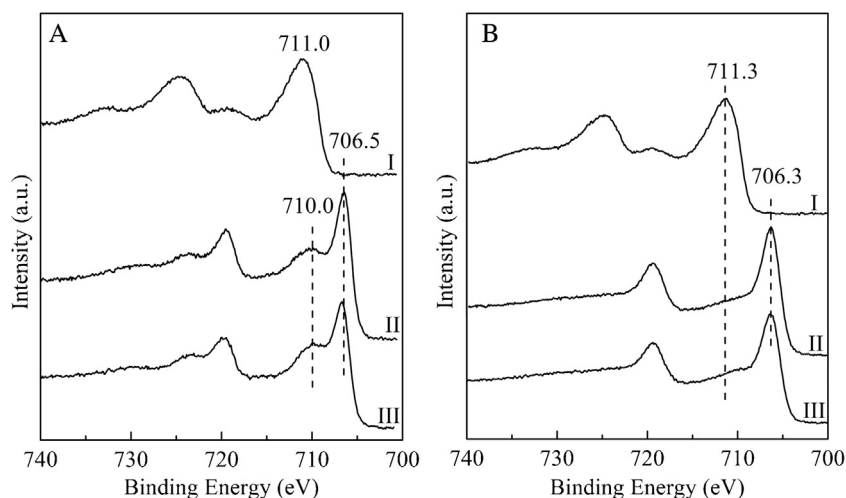


Fig. 9. Fe 2p XPS spectra of the (A) Fe/ α -Al₂O₃ and (B) S_{0.05}Fe/ α -Al₂O₃ catalysts after (I) treating under a flow of O₂/N₂ = 21/79 mixture (v/v, 50 mL/min) at 673 K for 20 min, (II) reducing under a flow of H₂ (50 mL/min) at 703 K for 120 min, (III) exposure to a flow of H₂/CO = 1 mixture (v/v, 10 mL/min) at 573 K for 60 min.

ever, the atomic ratio of S_{total}/Al decreased after the S_{0.05}Fe/ α -Al₂O₃ catalyst was reduced with H₂ at 703 K. This phenomenon could have resulted from the aggregation of iron and sulfur species during the H₂ reduction process as the atomic ratio of Fe/Al also decreased after H₂ reduction. Another possible reason for the decrease of surface S_{total}/Al ratio in the reduced catalyst is the release of sulfur species as gaseous SO₂ or H₂S during the reduction process. In consideration of the facts that no sulfide species was detected in the S 2p XPS peaks of a α -Al₂O₃ supported sulfate sample ((SO₄²⁻)/ α -Al₂O₃) reduced by H₂ at 703 K (Fig. S2), and that only the sulfide species was observed on a precipitated-iron catalyst after exposure to hydrogen at 673 K [23]. It is rational to infer that the sulfide species in the H₂-reduced S_xFe/ α -Al₂O₃ catalysts are located on the surface of metallic Fe particles. As can be seen from Fig. 9A, XPS spectrum of a freshly prepared Fe/ α -Al₂O₃ shows binding energy of 711.0 eV for the photoelectron peak of Fe 2p_{3/2}, which is typical for the Fe³⁺ species in the Fe₂O₃ [56,57]. After adding sulfur, the binding energy of Fe 2p_{3/2} increases slightly (711.3 eV). This may have resulted from the formation of Fe₂(SO₄)₃ as the binding energy of the Fe 2p_{3/2} XPS peak for the Fe³⁺ in Fe₂(SO₄)₃ (711.3 eV) is higher than that in Fe₂O₃ [58]. After reducing with H₂ at 703 K for 120 min, the XPS peak of Fe³⁺ (711.0 eV) decreased significantly in intensity. A broad peak with binding energy at 710.0 eV and a sharp peak with binding energy at 706.5 eV, assignable to Fe²⁺ [59] and metallic Fe species [57,59,60], respectively, were detected, indicating the reduction of Fe₂O₃ to FeO and Fe⁰. Comparing with Fe/ α -Al₂O₃, the Fe 2p_{3/2} XPS peak for Fe³⁺ in the H₂-reduced S_{0.05}Fe/ α -Al₂O₃ almost vanished, indicating that most of the Fe³⁺ species on the surface of sulfur modified catalyst was reduced to metallic Fe. This result is also in agreement with the results of H₂-TPR and *in situ* XRD characterizations shown above (Figs. 3 and 4), and provide further

evidence to show that sulfur promotes the reduction of Fe₂O₃ to metallic iron.

When H₂-reduced Fe/ α -Al₂O₃ sample was exposed to a mixture of CO/H₂ = 1 (v/v) at 573 K for 60 min, the XPS peak of Fe 2p_{3/2} at 706.5 eV shifted to 706.7 eV. The latter is in agreement with the Fe 2p_{3/2} XPS peak of iron carbide species [57]. This result indicates that metallic iron transforms to iron carbide under the CO/H₂ atmosphere. The formation of iron carbide was further confirmed by the observation of a C 1s peak at 283.0 eV assignable to carbide species [57,59–61] on the CO/H₂ treated Fe/ α -Al₂O₃ sample (Fig. S3). Comparing with the CO/H₂ pretreated sulfur free Fe/ α -Al₂O₃ catalyst, no significant shift in the Fe 2p_{3/2} XPS peak was observed on the CO/H₂ pretreated S_{0.05}Fe/ α -Al₂O₃ sample, indicating that most of the Fe species remained in metallic state under FTS reaction conditions. This is also confirmed by the C 1s XPS spectra of the sample (Fig. S3) in which no obvious XPS peak corresponding to the carbide species was observed on the CO/H₂ pretreated S_{0.05}Fe/ α -Al₂O₃. These results indicate that the addition of sulfur suppresses carburization of metallic Fe species.

Comparing with the H₂-reduced S_{0.05}Fe/ α -Al₂O₃ sample, no significant change in surface atomic ratios and binding energy of the elements was seen on the S_{0.05}Fe/ α -Al₂O₃ catalyst after the FTS reaction at 573 K for 1 h, indicating that the chemical state and surface concentration of the sulfur species were not significantly affected by syngas.

To investigate the effect of sulfur on the activity and selectivity of the α -Al₂O₃ supported iron catalyst for the FTS reaction, CO hydrogenation performance of the S_xFe/ α -Al₂O₃ catalysts with different S/Fe molar ratios was tested at 573 K and 2.0 MPa using H₂/CO = 1 as reaction feed, and the results are summarized in Table 2. It can be seen that sulfur modification in a range of S/Fe molar ratio between

Table 2
Catalytic performance of S_xFe/ α -Al₂O₃ catalysts for CO hydrogenation.

S/Fe (mol/mol)	Xco (%)	FTY (10 ⁻⁴ mol _{CO} /(s g _{Fe}))	Selectivity (%)				O/P ratio of C ₂ –C ₄	CO ₂ (% of CO converted)	α
			CH ₄	C ₂ –C ₄ olefins	C ₂ –C ₄ paraffins	C ₅ +			
0	49.3	7.0	18.0	28.9	16.3	36.7	1.8	39.2	0.58
0.001	39.6	5.6	18.9	26.5	20.7	33.9	1.3	32.2	0.57
0.002	26.4	3.7	20.6	26.4	23.0	29.7	1.1	24.6	0.54
0.005	17.4	2.5	17.8	26.7	24.1	31.5	1.1	17.4	0.52
0.01	5.6	0.8	17.2	25.8	29.5	27.6	0.9	13.0	0.48
0.02	–	–	–	–	–	–	–	–	–

Note: Reaction conditions: H₂/CO = 1, 2.0 MPa, 24000 mL/(h g_{cat}), 573 K, TOS = 7 h.

‘–’ indicates almost no activity.

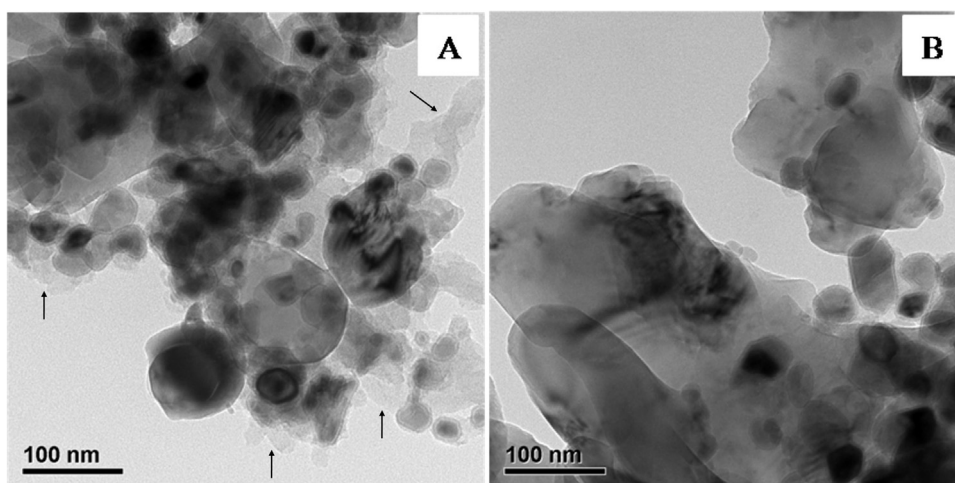


Fig. 10. TEM images of the (A) Fe/ α -Al₂O₃ and (B) S_{0.01}Fe/ α -Al₂O₃ catalysts after FTS reaction. (2.0 MPa, H₂/CO = 1, 573 K).

0.001 and 0.01 decreases the CO conversion. In the case of S_{0.02}Fe/ α -Al₂O₃ catalyst, the FTS activity almost diminished to zero. This result indicates that, for the FTS reaction at 573 K, a small amount of sulfur (S/Fe = 0.02) was sufficient to completely deactivate the Fe/ α -Al₂O₃ catalyst. As indicated in the XPS spectra of the catalysts after the FTS reaction (Fig. 8), both sulfate and sulfur species were detected on the S_{0.05}Fe/ α -Al₂O₃. It has been shown that the nature of sulfur species (S²⁻, SO₄²⁻) would affect catalyst behavior for the FTS reaction [13,23]. A low-level sulfide iron catalyst with sulfate species on the surface was associated with improving the FTS activity, whereas the presence of S²⁻ ions at the surface showed effects of poisoning [13]. It was also found that, at low sulfur coverage, each sulfide atom was able to poison 8–10 atoms of iron [62]. Based on these results, it can be concluded that the sulfide species is responsible for the lower FTS activity of the sulfur modified Fe/ α -Al₂O₃ catalysts as compared with the sulfur free one. The poisoning effect of sulfur on the Fe/ α -Al₂O₃ catalyst may have resulted from the blockage of catalytic metal sites that prevent the adsorption and activation of reactant molecules. As shown in Table 2, the presence of sulfur also had a profound impact on the product distribution of the FTS reaction. With increasing the S/Fe molar ratios in the catalysts from 0 to 0.01, the selectivity toward C₅₊ hydrocarbons and the chain growth probabilities (α) of the catalysts continuously decreased from 36.7 to 27.6 % and 0.58 to 0.48, respectively, whereas the total selectivity to C₂–C₄ hydrocarbons increased gradually from 45.2 to 55.3 %. Comparatively, CH₄ selectivity of the catalysts was not strongly affected by sulfur. As shown in Fig. S4, product distribution over Fe/ α -Al₂O₃ catalyst follows the ASF model, whereas CH₄ selectivities over S_{0.005}Fe/ α -Al₂O₃ and S_{0.01}Fe/ α -Al₂O₃ catalysts are below the ASF prediction, indicating that addition of sulfur resulted in a deviation of CH₄ selectivity from the ASF model. It is also worth to note that sulfur-modification decreases the olefin-to-paraffin (O/P) ratio of C₂–C₄ hydrocarbons in the reaction products. With increasing of the sulfur content (S/Fe molar ratio) in the catalysts from 0.001 to 0.01, the selectivity of C₂–C₄ olefins remained almost unchanged, whereas the selective of C₂–C₄ paraffins increased from 20.7 to 29.5%. This phenomenon can be well explained by that sulfur modification increasing the H/C ratio on the catalyst surface. As indicated in the H₂-TPD (Fig. 6) and CO-TPD (Fig. 7) profiles of the catalysts, the intensity of the CO desorption peak with T_m > 600 K corresponding to the dissociative adsorption of CO decreased with increasing of sulfur content, whereas the intensity of the H₂ desorption peak at temperature of about 500 K increased slightly with increasing of sulfur content. This will lead to an enhanced hydrogenation ability of the carbon

species on the catalyst surface. As a result, the chain growth termination by hydrogen and the olefin hydrogenation are promoted. The hydrocarbon products shift to short-chain molecules containing more paraffins. Higher H/C ratio leads to lower C₅₊ selectivity and O/P ratio are well documented in the literature [63].

Fig. 10 shows the TEM image of the Fe/ α -Al₂O₃ and S_{0.01}Fe/ α -Al₂O₃ catalysts after the FTS reaction at 573 K and 2.0 MPa. Comparing with the particle size of H₂-reduced samples (Fig. 2), a significant increase in the average size of iron-containing particles was observed on the spent Fe/ α -Al₂O₃ (25 ± 7 nm) and S_{0.01}Fe/ α -Al₂O₃ (30 ± 10 nm) catalysts. It can also be seen that the surface of spent Fe/ α -Al₂O₃ catalyst (Fig. 10A) is covered with large amounts of amorphous carbon (pointed by arrows). Comparatively, carbon deposition on the S_{0.01}Fe/ α -Al₂O₃ catalyst (Fig. 10B) is negligible. This indicates that the addition of sulfur decreased the formation of amorphous carbon. The study of Torres Galvis et al. also indicated

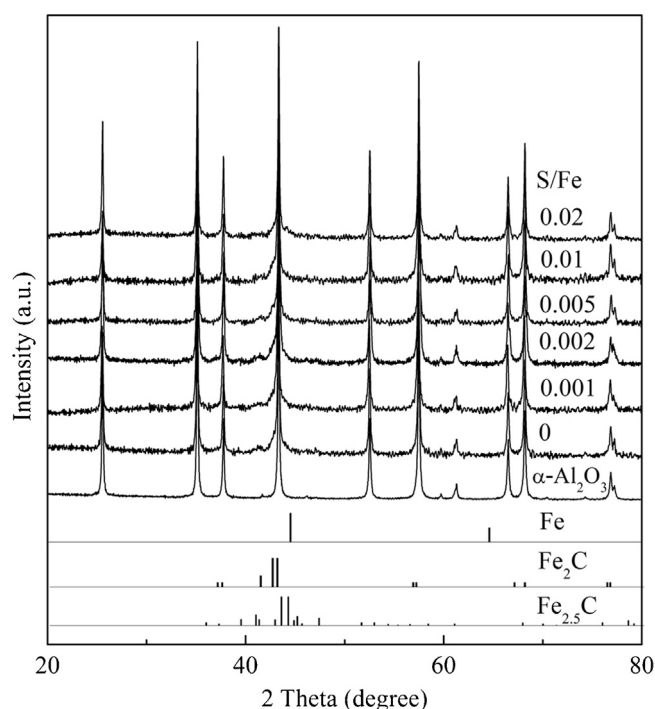


Fig. 11. XRD patterns of the S_xFe/ α -Al₂O₃ catalysts after the FTS reaction. (H₂/CO = 1, 2.0 MPa, 573 K).

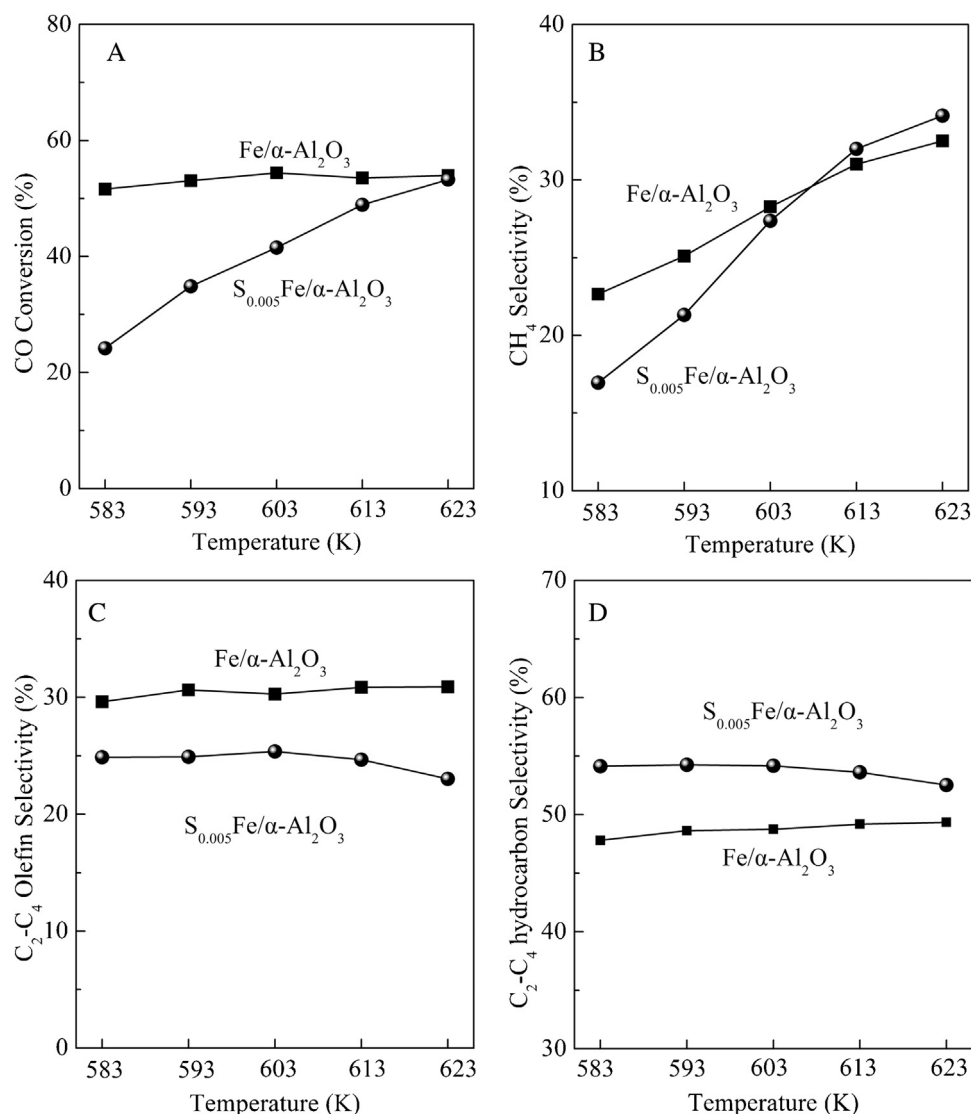


Fig. 12. Effect of reaction temperature on the (A) CO conversion, (B) CH₄ selectivity, (C) C₂–C₄ olefin selectivity and (D) C₂–C₄ hydrocarbon selectivity of the Fe/α-Al₂O₃ and S_{0.005}Fe/α-Al₂O₃ catalysts. (Reaction conditions: H₂/CO = 1, 2.0 MPa, 24000 mL/(h g_{cat}), TOS = 3 h.)

that the catalysts modified with sulfur show a low amount of carbon deposits [16]. This observation can also be related to suppression of dissociative activation of CO species by sulfur. XRD patterns of the Fe/α-Al₂O₃ and S_xFe/α-Al₂O₃ ($x = 0.001$ – 0.02) catalysts after FTS reaction are displayed in Fig. 11. It can be seen that XRD peaks attributed to iron carbides (Fe_xC_y) were observed on all catalysts, and no diffraction peak referring to metallic iron or iron oxides was detected. This observation is different from the results of XPS analysis of the CO/H₂ pretreated S_{0.05}Fe/α-Al₂O₃ sample (Fig. S3). The underlying reason for the difference may be that the sample for XRD analysis was treated under high pressure condition which speeded up the carbonization of metallic iron species in the catalyst. Since most intense peak of iron carbides partly overlaps with the peak of α-Al₂O₃ at about 43.5°, the type of iron carbides species formed on the catalysts can not be clearly identified by XRD. Nevertheless, the XRD patterns clearly indicate that the intensities of the diffraction peak of iron carbides (especially the peaks for Fe₂C) on the used catalysts gradually decrease with the increasing of S/Fe molar ratios. Bromfield et al. reported a similar observation for a precipitated iron catalyst modified by Na₂S [13]. This phenomenon can be understood in terms of sulfur suppressing the dissociative adsorption of CO, and thereby inhibiting the carburization of iron species.

Under FTS reaction conditions, the iron-based catalyst is known to be composed of many phases, including α-Fe, Fe₃O₄ and iron carbides [64]. Although there is considerable controversy concerning of the catalytically active iron phase as well as the role of each iron carbide species in the FTS reaction, the catalysts with more iron carbide species were often found to exhibit high activity for the FTS reaction [65–68]. This suggests that lower CO conversions observed on the S_xFe/α-Al₂O₃ as compared with that on Fe/α-Al₂O₃ may have resulted from the suppression of carburization of iron species by sulfur under FTS reaction conditions. As indicated in the CO-TPD profiles (Fig. 7) of the S_xFe/α-Al₂O₃ catalysts, addition of sulfur significantly decreases the amount of dissociative adsorbed CO on the Fe/α-Al₂O₃ catalyst, especially on the samples with S/Fe molar ratio above 0.01. This will result in a decrease in the concentration of active carbon species (C_{ad}) available for the formation of iron carbides and hydrocarbons, and eventually lead to a decline in FTS activity of the catalysts.

Fig. 12 shows the effect of reaction temperature on FTS performance of the Fe/α-Al₂O₃ and S_{0.005}Fe/α-Al₂O₃ catalysts. It can be seen that the addition of sulfur decreases the CO conversion of Fe/α-Al₂O₃ catalyst at low reaction temperature. As the reaction temperature was increased from 583 to 623 K, CO conversion of

Fe/ α -Al₂O₃ catalyst remains almost unchanged, while CO conversion of S_{0.005}Fe/ α -Al₂O₃ catalyst increases significantly. At 623 K, CO conversion of S_{0.005}Fe/ α -Al₂O₃ catalyst almost equals to that of Fe/ α -Al₂O₃. This observation indicates that the poisoning effect of sulfur on Fe/ α -Al₂O₃ catalyst is more significant at low reaction temperature. The significant difference in the dependence of activity (CO conversion) of Fe/ α -Al₂O₃ and S_{0.005}Fe/ α -Al₂O₃ catalysts on the reaction temperature could be related to the difference in the extent of carbon formation over the catalysts under FTS reaction conditions. As indicated in the TEM image of the spent Fe/ α -Al₂O₃ catalyst shown in Fig. 10, severe carbon deposition occurred on the Fe/ α -Al₂O₃ catalyst after the FTS reaction. It is well known that at high temperature, carbon formation by the Boudouard reaction will be more facile [69]. If the carbon species are not removed by hydrogenation to form hydrocarbons, they can block the active catalytic sites and eventually cause breakup of catalyst particles when they form carbon filaments [70–73]. Comparatively, the extent of carbon deposition on the sulfur modified Fe/ α -Al₂O₃ under FTS reaction conditions is much less significant (9–10). It could be the reason why the activity (CO conversion) of sulfur free catalyst is almost not affected by the variation of temperature, whereas the activity of sulfur modified catalysts increased significantly with increasing reaction temperature. This result indicates that the activity of iron catalyst poisoned by sulfur depends on temperature and thus the increasing of reaction temperature can improve the Fe/ α -Al₂O₃ catalyst's resistance to sulfur poisoning.

As also indicated in Fig. 12, the CH₄ selectivity of Fe/ α -Al₂O₃ and S_{0.005}Fe/ α -Al₂O₃ catalysts increased greatly with the increasing of reaction temperature. S_{0.005}Fe/ α -Al₂O₃ catalyst exhibited similar or slightly higher CH₄ selectivity in comparison with Fe/ α -Al₂O₃ catalyst at similar CO conversion level at 623 K. It indicates that, under the reaction conditions of this study, addition of sulfur does not reduce CH₄ selectivity of the catalysts. Within a temperature range between 583 and 623 K, the S_{0.005}Fe/ α -Al₂O₃ catalyst exhibited lower C₂–C₄ olefin selectivity and higher C₂–C₄ hydrocarbon selectivity as compared with Fe/ α -Al₂O₃ catalyst. These results indicate that both high reaction temperature and sulfur modification are in favor of the formation of lighter hydrocarbons. However, the olefin to paraffin (O/P) ratio of the C₂–C₄ fraction of S_{0.005}Fe/ α -Al₂O₃ is lower than that of Fe/ α -Al₂O₃, leading to a decrease in olefin selectivity over the sulfur modified catalyst.

4. Conclusions

The effects of sulfur (S/Fe = 0.001 ~ 0.05, molar ratio) on Fe/ α -Al₂O₃ catalysts for FTS were systematically studied. The catalysts were characterized by TEM, (*in situ*) XRD, XPS, and TPR/TPD. The results show that incorporation of sulfur into a Fe/ α -Al₂O₃ catalyst promoted the reduction of FeO to metallic Fe during the H₂ reduction process. With increasing the sulfur content, the rates of reduction, carburization and carbon deposition of the catalysts under CO atmosphere decreased. Sulfide (S²⁻) and sulfate species (SO₄²⁻) are the main sulfur components in the S_xFe/ α -Al₂O₃ catalysts after H₂ reduction and FTS reaction. The sulfide species on the H₂-reduced S_xFe/ α -Al₂O₃ catalysts are located on the surface of metallic Fe particles. A small amount of sulfur could enhance the amount of H₂ desorption from the catalyst surface at a temperature of about 500 K. However, an excess of sulfur on the catalyst suppressed the H₂ adsorption. The dissociative adsorption of CO on the H₂-reduced Fe/ α -Al₂O₃ catalysts was retarded by the addition of sulfur. The presence of sulfur inhibited the formation of iron carbides and amorphous carbon during FTS reaction. FTS performance of S_xFe/ α -Al₂O₃ depended on the content of sulfur as well as on the reaction temperature. The negative effect of sulfur on the FTS activity of Fe/ α -Al₂O₃ catalyst can be related to the inhibition of

CO dissociation and iron carbide formation over the catalyst under the FTS conditions (2.0 MPa, H₂/CO = 1, 573 K). Resistance to sulfur poisoning of the Fe/ α -Al₂O₃ catalyst was found to improve with increasing reaction temperature. The presence of sulfur in the catalyst also resulted in a decrease in selectivity of C₅₊ hydrocarbons as well as olefin to paraffin (O/P) ratio of C₂–C₄ products and promoted the formation of C₂–C₄ hydrocarbons. The higher surface H/C ratio over the sulfur modified catalyst under FTS conditions was proposed to be the main factor responsible for the changes in the product distribution.

Acknowledgments

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.apcata.2015.12.023>.

References

- [1] Q.H. Zhang, J.C. Kang, Y. Wang, *ChemCatChem* 2 (2010) 1030–1058.
- [2] H.M. Torres Galvis, K.P. de Jong, *ACS Catal.* 3 (2013) 2130–2149.
- [3] M. Janardanarao, *Ind. Eng. Chem. Res.* 29 (1990) 1735–1753.
- [4] R.J. Madon, H. Shaw, *Catal. Rev. Sci. Eng.* 15 (1977) 69–106.
- [5] J.F. Shultz, L.J.E. Hofer, F.S. Karn, R.B. Anderson, *J. Phys. Chem.* 66 (1962) 501–506.
- [6] J. Bayer, K.C. Stein, L.J.E. Hofer, R.B. Anderson, *J. Catal.* 3 (1964) 145–155.
- [7] F.S. Karn, J.F. Shultz, R.E. Kelly, R.B. Anderson, *I&EC Prod. Res. Dev.* 3 (1964) 33–38.
- [8] R.B. Anderson, F.S. Karn, J.F. Shultz, *J. Catal.* 4 (1965) 56–63.
- [9] P.K. Agrawal, W.D. Fitzharris, J.R. Katzer, *Stud. Surf. Sci. Catal.* 6 (1980) 179–200.
- [10] J. Barrault, C. Forquy, V. Perrichon, *Appl. Catal.* 5 (1983) 119–125.
- [11] C.H. Bartholomew, R.M. Bowman, *Appl. Catal.* 15 (1985) 59–67.
- [12] Z.T. Liu, J.L. Zhou, B.J. Zhang, *J. Mol. Catal.* 94 (1994) 255–261.
- [13] T.C. Bromfield, N.J. Coville, *Appl. Catal. A: Gen.* 186 (1999) 297–307.
- [14] J.A. Kritzinger, *Catal. Today* 71 (2002) 307–318.
- [15] T.Z. Li, Y. Yang, Z.C. Tao, H.J. Wan, X. An, C.H. Zhang, H.W. Xiang, Y.W. Li, *J. Nat. Gas Chem.* 16 (2007) 354–362.
- [16] H.M. Torres Galvis, A.C.J. Koeken, J.H. Bitter, T. Davidian, M. Ruitenbeek, A.I. Dugulan, K.P. de Jong, *J. Catal.* 303 (2013) 22–30.
- [17] A.J. McCue, J.A. Anderson, *Catal. Sci. Technol.* 4 (2014) 272–294.
- [18] H.G. Stenger, C.N. Satterfield, *I&EC Process Des. Dev.* 24 (1985) 415–420.
- [19] Z.T. Liu, Y.W. Li, J.L. Zhou, Z.X. Zhang, B.J. Zhang, *Appl. Catal. A: Gen.* 161 (1997) 137–151.
- [20] B. Wu, L. Tian, L. Bai, Z. Zhang, H. Xiang, Y.W. Li, *Catal. Commun.* 5 (2004) 253–257.
- [21] B.S. Wu, L. Bai, H.W. Xiang, Y.W. Li, Z.X. Zhang, B. Zhong, *Fuel* 83 (2004) 205–212.
- [22] W.L. van Dijk, J.W. Niemantsverdriet, A.M. van der Kraan, H.S. van der Baan, *Appl. Catal.* 2 (1982) 273–288.
- [23] T.C. Bromfield, N.J. Coville, *Appl. Surf. Sci.* 119 (1997) 19–24.
- [24] H.M. Torres Galvis, A.C.J. Koeken, J.H. Bitter, T. Davidian, M. Ruitenbeek, A.I. Dugulan, K.P. de Jong, *Catal. Today* 215 (2013) 95–102.
- [25] H.M. Torres Galvis, J.H. Bitter, C.B. Khare, M. Ruitenbeek, A.I. Dugulan, K.P. de Jong, *Science* 335 (2012) 835–838.
- [26] C.D. Wagner, L.E. Davis, M.V. Zeller, J.A. Taylor, R.H. Raymond, L.H. Gale, *Surf. Interface Anal.* 3 (1981) 211–225.
- [27] T.R. Motjope, H.T. Dlamini, G.R. Hearne, N.J. Coville, *Catal. Today* 71 (2002) 335–341.
- [28] J. Xu, C.H. Bartholomew, *J. Phys. Chem. B* 109 (2005) 2392–2403.
- [29] Z.H. Chonco, A. Ferreira, L. Lodya, M. Claeys, E. van Steen, *J. Catal.* 307 (2013) 283–294.
- [30] T.R. Motjope, T.H. Dlamini, *AIP Conf. Proc.* 765 (2005) 43–48.
- [31] H. Wang, Y. Yang, B.S. Wu, J. Xu, M.Y. Ding, H.L. Wang, W.H. Fan, H.W. Xiang, Y.W. Li, *J. Mol. Catal. A: Chem.* 308 (2009) 96–107.
- [32] S. Li, A. Li, S. Krishnamoorthy, E. Iglesia, *Catal. Lett.* 77 (2001) 197–205.
- [33] X. Cui, J. Xu, C. Zhang, Y. Yang, P. Gao, B. Wu, Y. Li, *J. Catal.* 282 (2011) 35–46.
- [34] M.C. Ribeiro, G. Jacobs, B.H. Davis, D.C. Cronauer, A.J. Kropf, C.L. Marshall, *J. Phys. Chem. C* 114 (2010) 7895–7903.

- [35] J. Benziger, R.J. Madix, *Surf. Sci.* 94 (1980) 119–153.
- [36] D.W. Moon, D.J. Dwyer, S.L. Bernasek, *Surf. Sci.* 163 (1985) 215–229.
- [37] D.W. Moon, S.L. Bernasek, J.P. Lu, J.L. Gland, D.J. Dwyer, *Surf. Sci.* 184 (1987) 90–108.
- [38] F. Bozso, G. Ertl, M. Grunze, M. Weiss, *Appl. Surf. Sci.* 1 (1977) 103–119.
- [39] P.B. Merrill, R.J. Madix, *Surf. Sci.* 347 (1996) 249–264.
- [40] H. Ahlafi, D. Bianchi, C.O. Bennett, *Appl. Catal.* 66 (1990) 99–110.
- [41] J. Li, C. Zhang, X. Cheng, M. Qing, J. Xu, B. Wu, Y. Yang, Y. Li, *Appl. Catal. A: Gen.* 464–465 (2013) 10–19.
- [42] G.D. Weatherbee, J.L. Rankin, C.H. Bartholomew, *Appl. Catal.* 11 (1984) 73–84.
- [43] Y. Amenomiya, G. Pleizier, *J. Catal.* 28 (1973) 442–454.
- [44] H.Z. Liu, C.B. Liu, X.N. Li, Y.Q. Cen, *Ind. Eng. Chem. Res.* 42 (2003) 1347–1349.
- [45] W. Arabczyk, I. Jasińska, R. Pelka, *Catal. Today* 169 (2011) 97–101.
- [46] G. Brodén, G. Gafner, H.P. Bonzel, *Surf. Sci.* 84 (1979) 295–314.
- [47] U. Seip, I.C. Bassignana, J. Küppers, G. Ertl, *Surf. Sci.* 160 (1985) 400–418.
- [48] U. Seip, M.C. Tsai, K. Christmann, J. Küppers, G. Ertl, *Surf. Sci.* 139 (1984) 29–42.
- [49] J.L. Rankin, C.H. Bartholomew, *J. Catal.* 100 (1986) 533–540.
- [50] J. Oh, T. Kondo, D. Hatake, K. Arakawa, T. Suzuki, D. Sekiba, J. Nakamura, *J. Phys. Chem. C* 116 (2012) 7741–7747.
- [51] G. Brodén, G. Gafner, H.P. Bonzel, *Appl. Phys.* 13 (1977) 333–342.
- [52] T.N. Rhodin, C.F. Brucker, *Solid State Commun.* 23 (1977) 275–279.
- [53] F. Jona, K.O. Legg, H.D. Shih, D.W. Jepsen, P.M. Marcus, *Phys. Rev. Lett.* 40 (1978) 1466–1469.
- [54] K.O. Legg, F. Jona, D.W. Jepsen, P.M. Marcus, *Surf. Sci.* 66 (1977) 25–37.
- [55] T. Jin, M. Machida, T. Yamaguchi, K. Tanabe, *Inorg. Chem.* 23 (1984) 4396–4398.
- [56] E. Paparazzo, *J. Electron. Spectrosc. Relat. Phenom.* 43 (1987) 97–112.
- [57] C.S. Kuivila, J.B. Butt, P.C. Stair, *Appl. Surf. Sci.* 32 (1988) 99–121.
- [58] D. Brion, *Appl. Surf. Sci.* 5 (1980) 133–152.
- [59] E. de Smit, M.M. van Schooneveld, F. Cinquini, H. Bluhm, P. Sautet, F.M.F. de Groot, B.M. Weckhuysen, *Angew. Chem. Int. Ed.* 50 (2011) 1584–1588.
- [60] H.P. Bonzel, G. Brodén, H.J. Krebs, *Appl. Surf. Sci.* 16 (1983) 373–394.
- [61] T. Grzybek, H. Papp, M. Baerns, *Appl. Catal.* 29 (1987) 351–359.
- [62] C.H. Bartholomew, P.K. Agrawal, J.R. Katzer, *Adv. Catal.* 31 (1982) 135–242.
- [63] G.P. Van der Laan, A. Beenackers, *Catal. Rev. Sci. Eng.* 41 (1999) 255–318.
- [64] E. de Smit, B.M. Weckhuysen, *Chem. Soc. Rev.* 37 (2008) 2758–2781.
- [65] W. Chen, Z.L. Fan, X.L. Pan, X.H. Bao, *J. Am. Chem. Soc.* 130 (2008) 9414–9419.
- [66] M.D. Shroff, D.S. Kalakkad, K.E. Coulter, S.D. Kohler, M.S. Harrington, N.B. Jackson, A.G. Sault, A.K. Datye, *J. Catal.* 156 (1995) 185–207.
- [67] J.D. Xu, K.T. Zhu, X.F. Weng, W.Z. Weng, C.J. Huang, H.L. Wan, *Catal. Today* 215 (2013) 86–94.
- [68] J.A. Amelse, J.B. Butt, L.H. Schwartz, *J. Phys. Chem.* 82 (1978) 558–563.
- [69] A.C.J. Koeken, H.M. Torres Galvis, T. Davidian, M. Ruitenbeek, K.P. de Jong, *Angew. Chem. Int. Ed.* 51 (2012) 7190–7193.
- [70] M.E. Dry, A.P. Steynberg, *Stud. Surf. Sci. Catal.* 152 (2004) 406–481.
- [71] C.H. Bartholomew, *Appl. Catal. A: Gen.* 212 (2001) 17–60.
- [72] K.P. De Jong, J.W. Geus, *Catal. Rev.* 42 (2000) 481–510.
- [73] J.R. Rostrup-Nielsen, J. Sehested, *Adv. Catal.* 47 (2002) 65–139.

Supporting Information

Effect of sulfur on α -Al₂O₃-supported iron catalyst for Fischer-Tropsch synthesis

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Table S1. Peak deconvolution results of the H₂-TPR profile of Fe/ γ -Al₂O₃ catalyst.

Catalysts	Peak areas ($\times 10^6$)		
	Peak 1(< 700 K)	Peak 2 (700-940 K)	Peak 3 (> 940 K)
Fe/ α -Al ₂ O ₃	1.2	2.2	2.1

Table S2. Peak deconvolution results of the CO-TPR profiles of S_xFe/ γ -Al₂O₃ catalysts.

Catalysts	Peak areas ($\times 10^{-8}$)	
	Peak 1	Peak 2
Fe/ α -Al ₂ O ₃	1.2	14.6
S _{0.001} Fe/ α -Al ₂ O ₃	1.0	12.7
S _{0.002} Fe/ α -Al ₂ O ₃	1.1	15.0
S _{0.005} Fe/ α -Al ₂ O ₃	0.9	8.8
S _{0.01} Fe/ α -Al ₂ O ₃	1.0	6.0
S _{0.02} Fe/ α -Al ₂ O ₃	1.1	4.5
S _{0.05} Fe/ α -Al ₂ O ₃	1.2	4.0

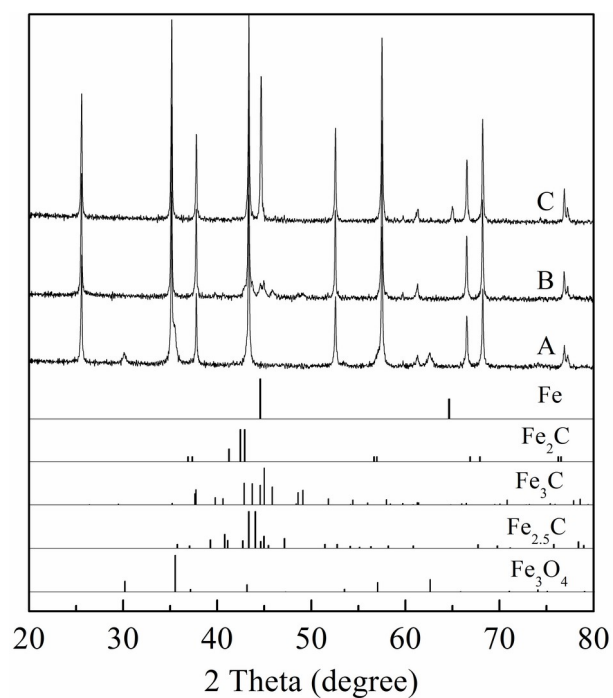


Fig. S1. XRD patterns recorded on the (A) Fe/ α - Al_2O_3 catalyst after treating in a flow of CO/He = 5/95 mixture (v/v, 50 mL/min) at 573 K, (B) Fe/ α - Al_2O_3 catalyst after treating in a flow of CO/He = 5/95 mixture (v/v, 50 mL/min) at 1073 K and (C) $\text{S}_{0.05}\text{Fe}/\alpha$ - Al_2O_3 catalyst after treating in a flow of CO/He = 5/95 mixture (v/v, 50 mL/min) at 1073 K.

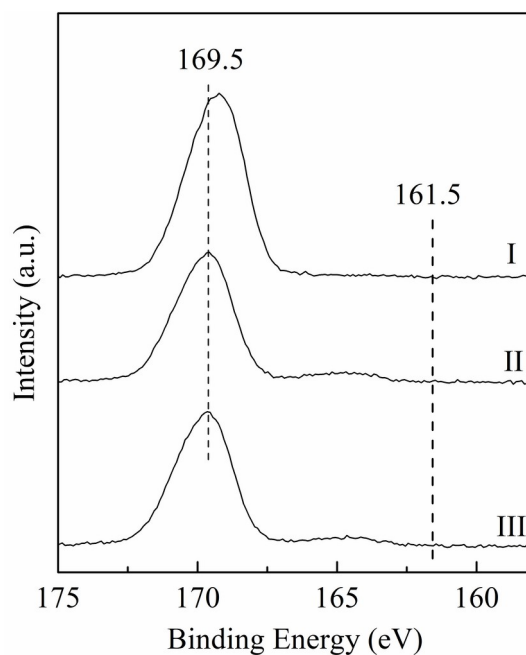


Fig. S2. S 2p XPS spectra of the 5wt% S/ α -Al₂O₃ catalyst prepared by heating the sample (I) in a flow of O₂/N₂ = 21/79 mixture (v/v, 50 mL/min) at 673 K for 20 min, (II) in a flow of H₂ (50 mL/min) at 703 K for 120 min, (III) in a flow of H₂/CO = 1 mixture (v/v, 10 mL/min) at 573 K for 60 min.

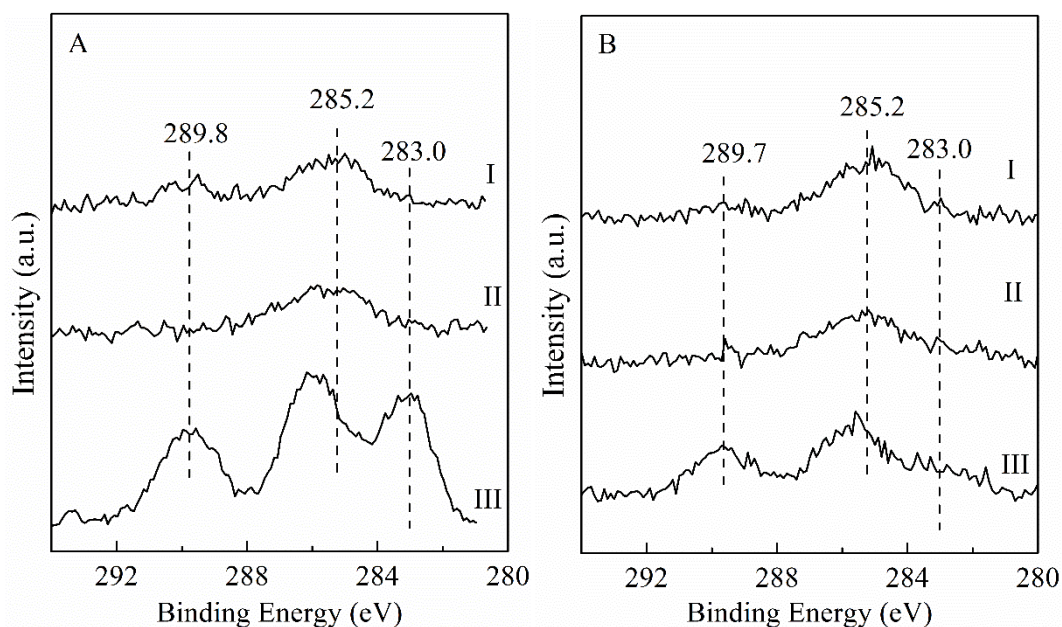


Fig. S3. C 1s XPS spectra of the (A) Fe/ α -Al₂O₃ and (B) S_{0.05}Fe/ α -Al₂O₃ catalysts prepared by heating the samples (I) in a flow of O₂/N₂ = 21/79 mixture (v/v, 50 mL/min) at 673 K for 20 min, (II) in a flow of H₂ (50 mL/min) at 703 K for 120 min,

(III) in a flow of $\text{H}_2/\text{CO} = 1$ mixture (v/v, 10 mL/min) at 573 K for 60 min.

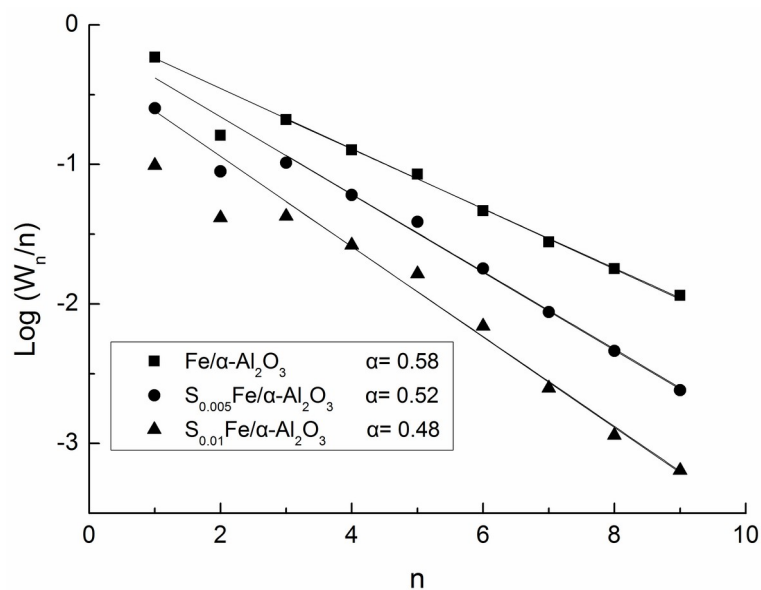


Fig. S4. Anderson-Schulz-Flory plots of the product distribution.

Reaction conditions: $\text{H}_2/\text{CO}=1$, 2.0 MPa, 24000 mL/(h · g_{cat}), 573 K, TOS = 7 h.